

MATH120 Discrete Mathematics

Tutorial 4

References

- Topic 5 and 6: Divisibility and Prime factorisation
- Sections 4.4, 4.10, 8.4 of Epp (5th Ed)
- Sections 4.3, 4.8, 8.4 of Epp (4th Ed)

Work on the following problems in groups during the tutorial in week 5.
Start with the marked questions to ensure good coverage of the material.

- ⇒ 1. [B] If $a, b, c \in \mathbb{Z}^+$ and $a|bc$, does it follow that $a|b$ or $a|c$? Prove or give a counterexample.
2. [B] Factorise 2020 in prime numbers. Use trial division to show that all the factors you write down are in fact primes.
3. [M] If $a, b \in \mathbb{Z}^+$ with $a = 630$, $\gcd(a, b) = 105$, and $\text{lcm}(a, b) = 242,550$; what is b ?
- ⇒ 4. [B] For each of the pairs $a, b \in \mathbb{Z}^+$,
- (a) $a = 231, b = 1820$
- (b) $a = 1369, b = 2597$
- do the following:
- (i) determine $\gcd(a, b)$,
- (ii) express $\gcd(a, b)$ as a linear combination of a, b ,
- (iii) find $\text{lcm}(a, b)$.
- ⇒ 5. [M] Prove the following statements.
- (a) For all $a, b \in \mathbb{Z}^+$, $a|b$ if and only if $\text{lcm}(a, b) = b$.
- (b) For $a, b, c \in \mathbb{Z}^+$, we have that $\gcd(\gcd(a, b), c) | \gcd(a, \gcd(b, c))$.
6. [B] Given the two integers 7, 114, 800 and 7, 882, 875:

- (a) write down both integers as a product of primes,
- (b) find the number of positive divisors for each integer,
- (c) determine their greatest common divisor,
- (d) determine their least common multiple.

$\Rightarrow$ 7. [M] Show the following.

- (a) If n is any positive integer whose last digit is a 5, then $5|n$.
- (b) An integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

8. (a) [D] What is a Mersenne prime?

- (b) [C] Let $m \in \mathbb{Z}^+$, $m > 1$. Show that $2^{2m} - 1$ is not a prime.

9. [C] Let $n!$ denote the product of the integers 1 through n .

- (a) What is the remainder when $(1! + 2! + 3! + 4! + 5! + 6! + \dots)$ is divided by 9?
- (b) Show that there exists a prime p such that $n < p \leq n! + 1$.